

ShopPilot

System Deep Dive — Architecture, Execution, Algorithms & Trade-offs

TechJam 2026 · Track 4 · Conversational Shopping Agent
Demo UI: Astrid CLI · Offline-first multi-turn shopping copilot

Metric (public 200, hash)	Value
TechnicalScore	~0.794
Hit@10	0.935
MRR	~0.558
MTTC	~3.03
vs weak BM25 starter Tech	~0.11 → ~0.794
Default path tokens / network	0 / offline

Compiled from implementation notes for study.
Repo: github.com/algorathem/techjam2026-shopping-copilot

1. What this system is

ShopPilot is a **headless, offline-first multi-turn shopping agent** over a frozen Amazon Clothing/Shoes/Jewelry catalog (~50k products). Each turn takes natural language, updates a compact dialogue state, hybrid-retrieves candidates, ranks them, and returns **Top-10 parent_asin** plus **exactly one** structured **ask_attribute** (or null). The official evaluator scores Hit@10, MRR, and MTTC into TechnicalScore.

It is **not** a chatbot UI product, not a required-LLM stack, and not a full neural DST. The demo face is the **Astrid** colored CLI; the project/metrics name is **ShopPilot**.

Problem the kit poses

- Vague multi-turn shoppers (browse, refine, override) vs one-shot keyword search
- Ambiguity: dress vs dress shoes; for my son vs women's items
- ≤ 10 turns or the session fails / MTTC hurts
- Must emit protocol-legal ask_attribute enum only
- Success = rank the **hidden target ASIN**, not “sound helpful”

Solution in one line

Messy multi-turn language → **SessionState** + hybrid FTS/dense retrieve + linear constraint rank + protocol-static clarify → right ASIN in fewer turns, offline.

2. Architecture (multi-turn, not one-shot)

Every user turn runs on **accumulated past state plus the new utterance**. Weak BM25 starter mostly uses tokens from the current message only. ShopPilot is DST-style state-update: retrieve and ask read the full SessionState.

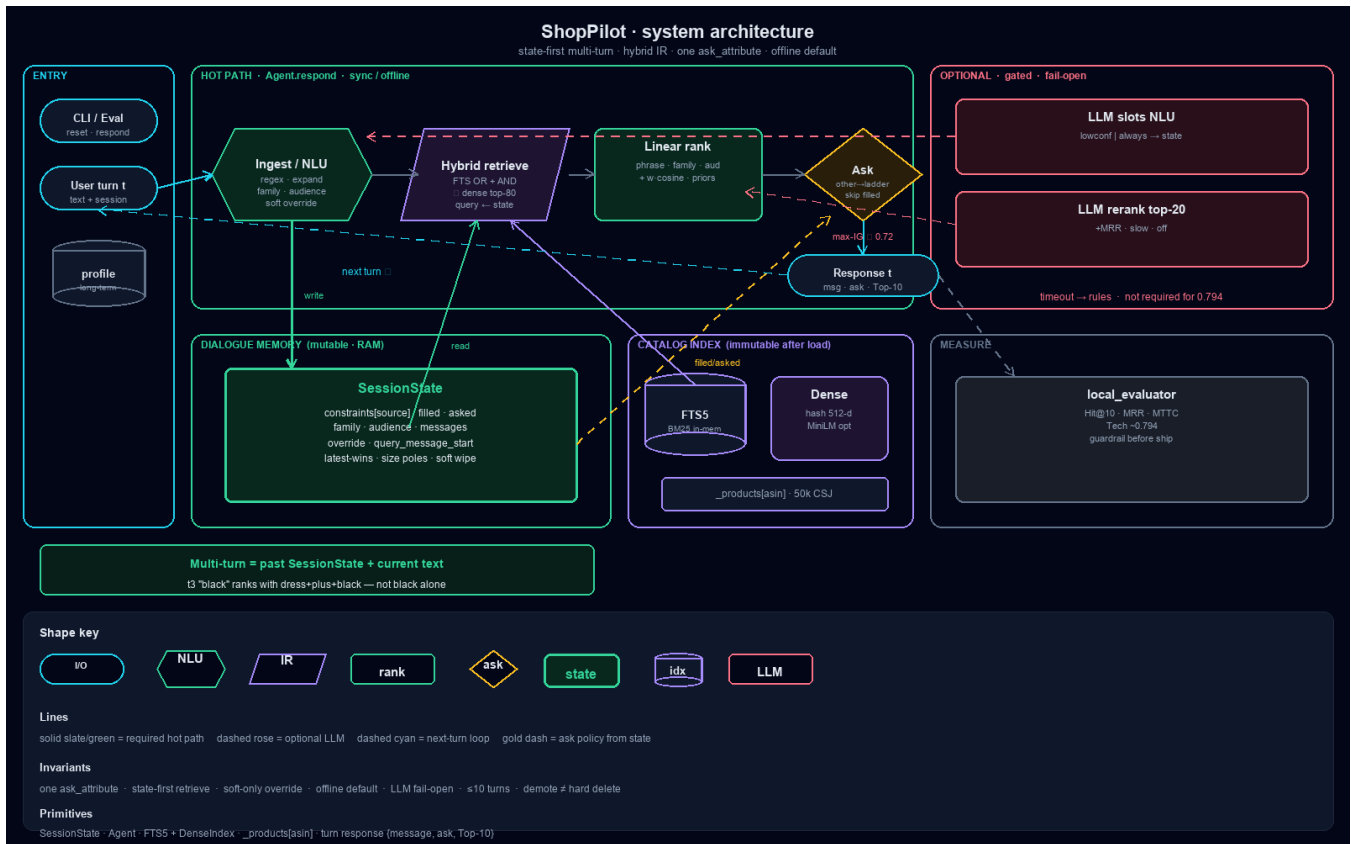


Figure: shape-coded architecture — stadium=I/O, hexagon=NLU, parallelogram=retrieve, diamond=ask, document=SessionState, cylinder=indexes, dashed rose=optional LLM.

Shape key

Shape	Aspect
Stadium (pill)	Turn I/O — user / response / CLI
Hexagon	NLU / ingest (parse, override, family, audience)
Parallelogram	Hybrid retrieve (FTS ∪ dense; query ← state)
Rounded rect	Linear rank / Dense / LLM / evaluator
Diamond	Ask decision (other → ladder; max-IG rejected)
Document	SessionState (mutable multi-turn memory)
Cylinder	Stores — FTS5, profile
Dashed rose zone	Optional LLM (fail-open, off by default)

Per-turn loop

```

respond(session_id, user_message, turn, top_k)
1. Load SessionState for session_id          # PAST state
2. messages.append(user_message)
3. _ingest → write slots/family/audience/override
4. _maybe_llm_slots (optional; usually no-op)
5. _retrieve → FTS + dense + _score            # query FROM full state
6. _next_ask → one ask_attribute | null
7. _compose_message
8. return {message, ask_attribute, recommendations, usage}

```

next turn (same session_id) reuses updated state until reset / turn 10

Example: t1 “dress” → t2 “plus size” → t3 “black” ranks with **dress + plus + black**, not “black” alone.

3. Core primitives

Primitive	Where	Meaning
SessionState	starter/agent.py	Compact DST: slots, sources, family, audience, history slice, override flags
Agent	starter/agent.py	Owns catalog index + sessions; public API reset / respond only
Product + FTS	Agent._products + sqlite FTS5	Catalog atom: parent_asin → text/facets/price; inverted index
DenseIndex	starter/dense.py	Semantic recall: hash 512-d n-grams or MiniLM (opt-in)
Turn response	respond() return dict	Protocol atom: message + one ask_attribute + Top-K ASINs

SessionState fields (domain memory)

- **constraints + constraint_sources** (soft | disclosed | override)
- **filled / dont_care / asked** — slot machine for clarify policy
- **category, product_family, audience** (family/audience are internal, not kit asks)
- **messages, query_message_start, discarded_terms** — history window after override
- **profile** — long-term prior from reset; **browsing**, token counters

Lifecycle & mutation

Primitive	Init	Who mutates	Persist / GC
SessionState	Agent.reset(session_id, profile)	respond → _ingest, ask append, optional LLM	RAM only; reset overwrites; process exit drops
Agent / catalog	Agent(catalog_path) once	Load only; catalog file never written	In-mem FTS + dense die with process
DenseIndex	DenseIndex.build at Agent init	Read-only at query time	Optional MiniLM .npz cache under data/
Turn response	End of each respond	Ephemeral; CLI may display-enrich titles	GC after caller; metrics may hit results.json

4. Exact execution lifecycle of one turn

Primary operation: **Agent.respond(session_id, user_message, turn, top_k)**. Entry surfaces: **cli_chat.py** (Astrid demo) or **evaluator/local_evaluator.py** (official score). Same agent API.

File-by-file order

#	File	Symbol	Role
0	cli_chat / local_evaluator	main	Args, env, construct Agent
1	starter/agent.py	Agent.__init__	Load catalog, FTS, dense
2	starter/agent.py	Agent.reset	New SessionState
3	starter/agent.py	Agent.respond	Turn entry
4	starter/agent.py	Agent._ingest	Rules NLU → write state
5	starter/llm_slots.py	_maybe_llm_slots path	Optional JSON NLU (off)
6	starter/agent.py	Agent._retrieve	FTS + dense + merge
7	starter/agent.py	Agent._score	Linear feature rank
8	starter/llm_rerank.py	rerank	Optional top-K reorder (off)
9	starter/agent.py	Agent._next_ask	one ask null
10	starter/agent.py	Agent._compose_message	NL message
11	—	return dict	Exit to CLI or evaluator

Payload transforms (boundaries)

- **In:** session_id, user_message, turn, top_k
- **After ingest:** category/family/audience/constraints/filled updated; override may soft-wipe
- **After retrieve:** ranked list[(asin, score)] from FTS∪dense + _score (+ optional dense weight)
- **After next_ask:** ask_attribute ∈ kit enum or null; asked[] grows
- **Out:** {message, ask_attribute, recommendations[{parent_asin, score}], usage}

Sync vs async

The entire default path is **synchronous, in-process, blocking**. No threads, queues, or asyncio. Optional Gemini calls are still **sync HTTP inside respond** (timeout → fail-open to rules). There is no background job system.

5. Algorithms & techniques

5.1 Hybrid retrieval (late fusion)

- **Sparse:** SQLite FTS5 — OR lane (high recall, ≤45 terms, LIMIT 220) + AND lane on top constraints (LIMIT 80); BM25 field weights title-heavy
- **Dense:** hash char 3-grams → 512-d signed buckets (default) or MiniLM embeddings (opt-in, cached)
- **Fusion:** final = lexical_score(...) + w_backend × cosine; w_hash=4.5, w_minilm=10
- Dense also pulls candidates FTS missed (top 80) and re-scores FTS set

5.2 Linear constraint rerank (_score)

- Phrase/title hits, token coverage of constraints
- Product family +6.5 hit / -8.0 miss (dress ≠ dress sandals)
- Audience +4.0 / -5.5 (for my son ≠ women's) — demote, don't hard-delete
- Profile tags + rating priors scaled by cold/warm/hot session specificity
- Soft budget distance; stars/n_ratings as tie-break

5.3 Slot NLU (rules-first)

- Regex: looking-for, what-matters-is, override phrases, no-preference
- expand_constraint_phrases → multi-slot fill in one freeform line
- classify_constraint → official slots; vibes→style; size cues (big enough)
- Latest-wins short soft color/material; plus↔petite size poles
- Kit disclosed feature sentences kept **whole** (atomizing them hurt Tech)

5.4 Intent override (soft-only wipe)

```

apply_override():
  drop constraints where source == "soft"
  keep disclosed / override facts
  clear asked, dont_care
  query_message_start = last message index
  discarded_terms ← tokens of dropped soft prefs # block FTS leak

```

5.5 Clarifying-question policy

Policy	Status	Notes
other first (if open)	SHIPPED	Simulator can reveal ≤2 any-type constraints
Static ASK_ORDER + skip filled/dont_care/asked	SHIPPED	Stable, protocol-fit
Stop asks near turn ≥ 9	SHIPPED	Leave headroom before hard fail
Pool max info-gain / entropy ask	REJECTED	Implemented but Tech ~0.72
Coverage-gated pool swap	OFF	Code retained; not score path

Facet entropy code remains for **message grounding** (corpus-grounded CQ spirit), not for choosing ask_attribute. Literature (Qulac, Bayesian IG, corpus CQ, DST) informs design; kit A/B decides what ships.

5.6 Optional LLM sidecars

Mode	Flag	Role	Default
Slots NLU always	SHOPPILOT_LLM_SLOTS=always	JSON parse every turn	off
Slots NLU lowconf	SHOPPILOT_LLM_SLOTS=lowconf	Only if rule confidence low	off
Rerank	SHOPPILOT_LLM_RERANK=1	Reorder top-20 titles	off

Needs SHOPPILOT_LLM=1 + API key. Timeouts fall back to rules. Measured Gemini rerank: tiny Tech bump (~+0.003), ~10× slower. Slot NLU often timed out here → same answers as rules. **Do not make LLM required for submit.**

6. Optimization tricks

- Everything hot-path in RAM: sqlite :memory: FTS, product dict, dense matrix
- Catalog loaded once per process; turns don't re-read JSONL
- FTS candidate caps (220/80) then Python score — not full 50k scan

- Hash dense 512×50k float32 ≈ 100MB; NumPy matmul cosine
- MiniLM embeddings cached compressed under data/
- Query term cap ~45; phrase early-continue in _score
- Stdlib-first path (dense=None without NumPy); optional deps only
- LLM timeouts caught — fail-open so hung API doesn't kill the turn

Typical offline turn is usually well under 100ms locally. Optional LLM adds seconds.

7. Explicit trade-offs

Choice	Won	Lost / cost
Static ask + other-first	Tech ~0.79; stable; protocol-fit	No adaptive “smart” questions
Max-IG asks	Theory-pretty	Tech ~0.72 — rejected
Hash hybrid dense	+Hit/MRR, offline	~100MB RAM; NumPy soft-dep
Rules NLU + tiny lexicons	Controllable, kit-safe	Miss open slang without LLM
Soft demote (family/audience)	Fewer false negatives	Some wrong items still lower-ranked
Hard eliminate missing tokens	Aggressive filter	Killed true ASINs — rejected
Offline default	Reproducible, feasible, 0 tokens	Weaker open-ended NLU than always-on LLM
Append-only turns, no locks	Simple	Not idempotent; not multi-writer safe

Meta-objective: public TechnicalScore under the frozen simulator beats abstract conversational-AI elegance.
Measure → keep or reject.

8. System invariants

Protocol (organizer)

- respond always returns message + ask_attribute + recommendations
- ask_attribute is one enum string or null — never a list; never invent gender
- Recommendations identified by parent_asin
- Useful work in ≤ 10 turns
- reset must precede respond

Dialogue state

- Ask field $\subseteq$ official enum; family/audience stay internal
- Don't re-ask filled $\cup$ dont_care $\cup$ asked (until override clears asked)
- Override wipes soft only; disclosed survives; discarded terms blocked in FTS
- Retrieve is state-first, not current-message-only
- Exclusive soft slots: latest-wins color/material; size poles

Catalog / ranking / runtime

- Catalog read-only; closed world of loaded ASINs
- Offline default; LLM fail-open
- Prefer demote over hard delete on sparse titles
- Single-threaded assumption; turn is fully sync
- respond is NOT idempotent if called twice with same text

9. Slots, family, audience (who owns what)

Concept	Owner	Notes
ask_attribute enum	Organizer kit	category, material, color, size, style, brand, budget, feature, use_case, other, null
ASK_ORDER / fill policy	ShopPilot	other-first then static ladder; skip filled
product_family	ShopPilot (internal)	dress/footwear/... — fixes CSJ path ambiguity
audience	ShopPilot (internal)	boys/men/... from gift phrases like for my son
Vibe words (cool/cute/...)	ShopPilot	Map to style slot; soft FTS text

Judges score product findability (Hit/MRR/MTTC), not whether you invented internal taxonomies. Family/audience matter only if they raise metrics or demo quality.

10. Why offline-first

- Reliability: no timeout/rate-limit kills a scored session
- Feasibility: kit does not require paid LLM or vector DB
- Latency & cost: sub-second turns, zero tokens on default
- Privacy / merchant-owned brain: catalog + chat stay local
- Determinism: same inputs $\rightarrow$ same Tech for A/B
- Graceful upgrade: rules spine + optional MiniLM + optional Gemini

11. Key files cheat sheet

Path	Responsibility
starter/agent.py	Orchestrator: SessionState, ingest, retrieve, score, ask
starter/dense.py	Dense hash / MiniLM lane
starter/rewrite.py	Brief query string for optional rerank
starter/llm_slots.py	Optional dual-meaning slot NLU
starter/llm_rerank.py	Optional top-K LLM rerank
cli_chat.py	Astrid demo CLI
evaluator/local_evaluator.py	Official Hit/MRR/MTTC loop
data/catalog.jsonl	Frozen product corpus
docs/architecture_diagram.png	Shape-coded architecture figure

Reproduce metrics

```
cd techjam2026-shopping-copilot
export SHOPPILOT_DENSE=hash
unset SHOPPILOT_LLM SHOPPILOT_LLM_SLOTS SHOPPILOT_LLM_RERANK
python3 -m unittest tests.test_agent_slots tests.test_dense -q
python3 -m evaluator.local_evaluator
python3 cli_chat.py --dense hash
```

12. Glossary

Term	Meaning here
DST	Dialogue state tracking — compact slots across turns
FTS5	SQLite full-text search (sparse BM25-like retrieve)
Hybrid IR	Sparse + dense recall, late score fusion
Hit@10	Target parent_asin in top 10 recommendations
MRR	Mean reciprocal rank of the target
MTTC	Mean turns to first correct hit (lower better)
TechnicalScore	$0.5 \cdot \text{Hit} + 0.3 \cdot \text{MRR} + 0.2 \cdot \text{Efficiency}(\text{MTTC})$
ask_attribute	Single kit enum field requesting a slot (or null)
other	Catch-all ask; simulator may dump multiple constraints
soft / disclosed	Constraint provenance for override hygiene
Fail-open LLM	On API failure, keep rule-based state and continue

13. Mental model (memorize this)

One primary operation = synchronous **Agent.respond**: load multi-turn **SessionState**, rule-ingest (optional LLM enrich) the new utterance, hybrid-retrieve and score from the **full state**, pick **one** protocol ask_attribute, compose a message, return Top-K ASINs — no background workers, offline by default, optional LLM is a gated sidecar.

For judges: offline hybrid shopping agent that lifts kit Tech $\sim 0.11 \rightarrow \sim 0.794$ with multi-turn state, override hygiene, and protocol-safe clarification — not a fine-tuned foundation model and not a required cloud API.